

REASON

Consider the transformation of a parent function translated by (h, k) so that $g(x) = f(x + h) + k$. With exponential functions, the value of k dictates the shift of the horizontal asymptote. With logarithmic functions, the value of h determines the shift of the asymptote.

EXAMPLE 2 Graph Transformations of Logarithmic Functions

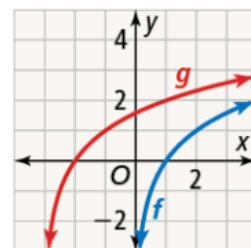
Graph the function $g(x) = \log_2(x + 3)$. How do the asymptote and x -intercept of the given function compare to those of the parent function $f(x) = \log_2 x$?

$$f(x) = \log_2 x$$

$$g(x) = \log_2(x + 3) = f(x - (-3))$$

Since $g(x) = f(x + 3)$, then g is the logarithmic function translated 3 units to the left.

The vertical asymptote and the x -intercept each shift 3 units to the left.



Try It! 2. Describe how each graph compares to the graph of $f(x) = \ln x$.

a. $g(x) = \ln x + 4$

b. $h(x) = 5 \ln x$

CONCEPTUAL UNDERSTANDING**LOOK FOR RELATIONSHIPS**

$f(x)$ is a translation of the parent function $y = 10^x$ one unit left.
 $f^{-1}(x)$ is a translation of the parent function $y = \log x$ one unit down. Graphical translations of a function and its inverse are directly related. Explain.

EXAMPLE 3 Inverses of Exponential and Logarithmic Functions

What is the equation of the inverse of the functions?

A. $f(x) = 10^{x+1}$

$$y = 10^{x+1}$$

$$x = 10^{y+1}$$

$$y + 1 = \log x$$

$$y = \log x - 1$$

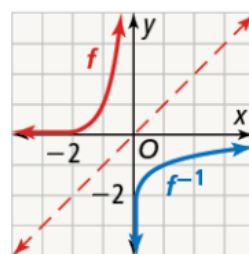
Rewrite the function as $y =$, interchange x and y , and solve for y .

The equation of the inverse of $f(x) = 10^{x+1}$ is $f^{-1}(x) = \log x - 1$.

Verify by function composition.

$$\begin{aligned} f(f^{-1}(x)) &= 10^{(\log x - 1) + 1} \\ &= 10^{\log x} \\ &= x \end{aligned}$$

$$\begin{aligned} f^{-1}(f(x)) &= \log 10^{x+1} - 1 \\ &= x + 1 - 1 \\ &= x \end{aligned}$$



B. $g(x) = \log_7(x + 5)$

$$y = \log_7(x + 5)$$

$$x = \log_7(y + 5)$$

$$y + 5 = 7^x$$

$$y = 7^x - 5$$

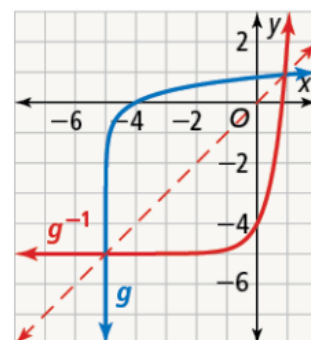
Rewrite as $y =$.

The equation of the inverse of $g(x) = \log_7(x + 5)$ is $g^{-1}(x) = 7^x - 5$.

Verify by function composition.

$$\begin{aligned} g(g^{-1}(x)) &= \log_7(7^x - 5 + 5) \\ &= \log_7 7^x \\ &= x \end{aligned}$$

$$\begin{aligned} g^{-1}(g(x)) &= 7^{\log_7(x + 5)} - 5 \\ &= x + 5 - 5 \\ &= x \end{aligned}$$



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Try It! 3. Find the inverse of each function. Then verify they are inverses by composition.

a. $f(x) = 3^{x+2}$

b. $g(x) = \log_7 x - 2$

EXAMPLE 4 Use a Logarithmic Function as a Model

A logarithmic function can approximate the altitude of a plane over time.

A. What is the altitude of the plane 10 minutes after takeoff?

Substitute 10 for t in the equation of the function.

$$\begin{aligned} A &= 9,200 \ln 10 + 10,000 \\ &\approx 9,200(2.303) + 10,000 \\ &\approx 21,188 + 10,000 \\ &\approx 31,188 \end{aligned}$$

After 10 minutes, the plane's altitude is about 31,188 feet.

B. How long after takeoff will the plane reach an altitude of 35,000 feet?

Rearrange the equation to isolate the variable t .

$$\begin{aligned} A &= 9,200 \ln t + 10,000 \\ A - 10,000 &= 9,200 \ln t \\ \frac{A - 10,000}{9,200} &= \ln t \\ t &= e^{\left(\frac{A - 10,000}{9,200}\right)} \end{aligned}$$

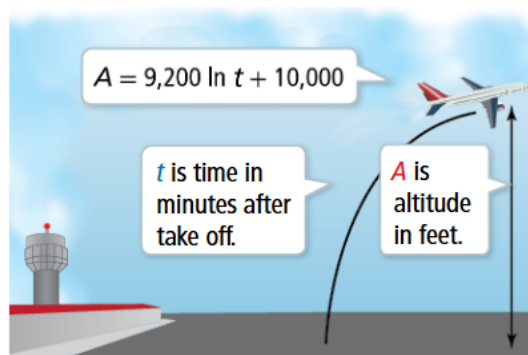
In the rearranged equation, time is a function of altitude, so use substitution to find time.

$$\begin{aligned} t &= e^{\left(\frac{35,000 - 10,000}{9,200}\right)} \\ &\approx 15.14 \end{aligned}$$

The plane will reach an altitude of 35,000 feet after about 15.14 minutes.

C. Evaluate the model against the altitude of the plane.

On the domain interval $(0, 0.337)$ the logarithmic function is negative. The altitude of the plane is not correct during this interval. Also, the plane starts at the origin, which is on the asymptote of the model.



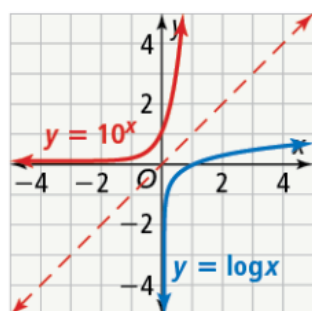
STUDY TIP

The original equation is a logarithmic function, while the rearranged equation is an exponential function. The exponential function is the inverse of the logarithmic function.

Try It! 4. Commercial airplanes usually fly no higher than 38,000 ft. How long will it take the plane to reach this altitude? How does a maximum cruising height compare with the model?

CONCEPT SUMMARY Logarithmic Functions

GRAPH



The functions are inverses so their graphs are reflections of each other across the line with equation $y = x$.

EQUATIONS

$$y = \log x$$

$$y = 10^x$$

KEY FEATURES

Domain: $\{x \mid x > 0\}$

Domain: *all real numbers*

Range: all real numbers

Range: $\{y \mid y > 0\}$

x-intercept: 1

y-intercept: 1

Asymptote: y-axis

Asymptote: x-axis

END BEHAVIOR

As $x \rightarrow 0$, $y \rightarrow -\infty$

As $x \rightarrow -\infty$, $y \rightarrow 0$

As $x \rightarrow \infty$, $y \rightarrow \infty$

As $x \rightarrow \infty$, $y \rightarrow \infty$

Do You UNDERSTAND?

- ESSENTIAL QUESTION** How is the relationship between logarithmic and exponential functions revealed in the key features of their graphs?
- Error Analysis** Raynard claims the domain of the function $y = \log_3 x$ is all real numbers. Explain the error Raynard made.
- Communicate Precisely** How are the graphs of $f(x) = \log_5 x$ and $g(x) = -\log_5 x$ related?
- Use Structure** Explain the necessary steps to find the inverse of $h(x) = \log_6 (x + 4)$

Do You KNOW HOW?

- Graph the function $y = \log_4 x$ and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior.
- Write the equation for the function $g(x)$, which can be described as a vertical shift $1\frac{1}{2}$ units up from the function $f(x) = \ln x - 1$.
- The function $y = 5 \ln(x + 1)$ gives y , the number of downloads, in hundreds, x minutes after the release of a song. Find the equation of the inverse and interpret its meaning.



- Sketch the functions represented by the tables. Identify which graph is the logarithmic function. Are the two functions inverses?

a.

x	10^{-10}	0.01	1	2	10
$g(x)$	-8	0	2	2.301	3

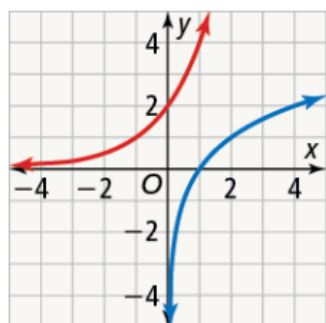
b.

x	-1	0	1	2	3	4
$h(x)$	0.001	0.01	0.1	1	10	100



UNDERSTAND

9. **Look for Relationships** Are the logarithmic and exponential functions shown inverses of each other? Explain.



10. **Communicate Precisely** How is the graph of the logarithmic function $g(x) = \log_2(x - 7)$ related to the graph of the function $f(x) = \log_2 x$? Explain your reasoning.

11. **Error Analysis** Describe and correct the error a student made in finding the inverse of the exponential function $f(x) = 5^{x-6} + 2$.

$$y = 5^{x-6} + 2$$

$$x = 5^{y-6} + 2$$

$$x - 2 = 5^{y-6}$$

$$y - 6 = \log_5 x - 2$$

$$y = \log_5 x - 2 + 6$$

$$y = \log_5 x + 4$$

$$f^{-1}(x) = \log_5 x + 4$$

Write in $y = f(x)$ form.

Interchange x and y .

Subtract 2 from each side.

Rewrite in logarithmic form.

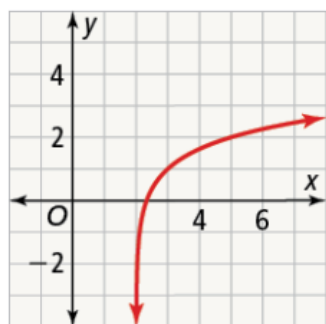
Add 6 to each side.

Simplify.



12. **Make Sense and Persevere** The number of members m who joined a new workout center w weeks after opening is modeled by the equation $m = 1.6^{w+2}$, where $0 \leq w \leq 10$. Find the inverse of the function and explain what the inverse tells you.

13. **Use Structure** The graph shows a transformation of the parent graph $f(x) = \log_3 x$. Write an equation for the graph.



PRACTICE

Graph each function and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior. SEE EXAMPLE 1

14. $y = \log_5 x$

15. $y = \log_8 x$

16. $y = \log_{\frac{3}{10}} x$

17. $y = \log_{0.1} x$

Describe the graph in terms of transformations of the parent function $f(x) = \log_6 x$. Compare the asymptote and x -intercept of the given function to the parent function. SEE EXAMPLE 2

18. $g(x) = \frac{1}{2} \log_6 x$

19. $g(x) = \log_6(-x)$

20. Describe how the graph of $g(x) = -\ln(x + 0.5)$ is related to the graph of $f(x) = \ln x$. SEE EXAMPLE 2

Find the equation of the inverse of each function. SEE EXAMPLE 3

21. $f(x) = 5^{x-3}$

22. $f(x) = \left(\frac{1}{2}\right)^{x-1}$

23. $f(x) = 6^{x+7}$

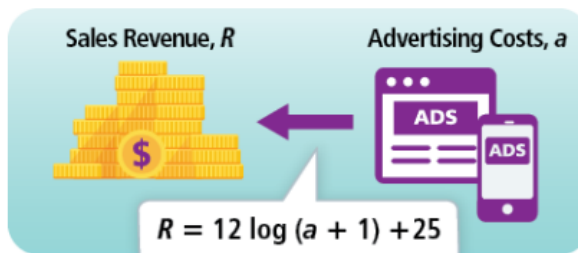
24. $f(x) = \log_2(8x)$

25. $f(x) = \ln(x + 3) - 1$

26. $f(x) = 4 \log_2(x - 3) + 2$

27. The altitude y , in feet, of a plane t minutes after takeoff is approximated by the function $y = 5,000 \ln(.05t) + 8,000$. Solve for t in terms of y . What is a situation in which it would be easier to use your new equation rather than the original? SEE EXAMPLE 4

28. A company uses this function to relate sales revenue, R , and advertising costs, a . What is the equation of the inverse of the equation? Which equation would be easier to use to find a value of a for a particular value of R ? SEE EXAMPLE 4



PRACTICE & PROBLEM SOLVING

APPLY

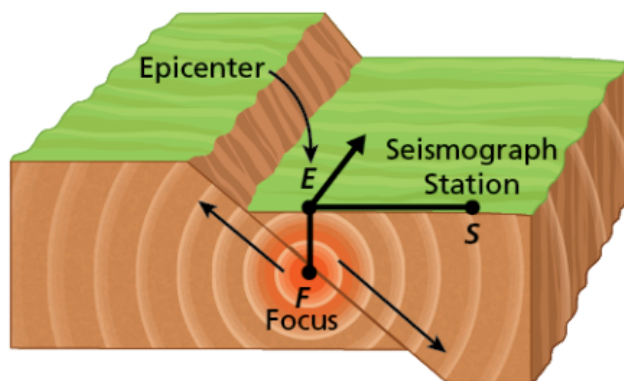
29. **Model with Mathematics** The equation $r = 90 - 25 \log(t + 1)$ is to model a student's retention r after taking a physics course where r represents a student's test score (as a percent), and t represents the number of months since taking the course.

- Make a table of values for ordered pairs that represent $r = 90 - 25 \log(t + 1)$, rounding to the nearest tenth. Then sketch the graph of the function on a coordinate plane through those ordered pairs. (You may use a graphing calculator to check.)
- Find the equation of the inverse. Interpret the meaning of this function.

30. **Higher Order Thinking** As shown by the diagram, an earthquake occurs below Earth's surface at point F (the focus). Point E , on the surface above the focus, is called the *epicenter*. A seismograph station at point S records the waves of energy generated by the earthquake. The surface wave magnitude M of the earthquake is given by this formula:

$$M = \log\left(\frac{A}{T}\right) + 1.66(\log D) + 3.3$$

In the formula, A is the amplitude of the ground motion in micrometers, T is the period in seconds, and D is the measure of ES in degrees.



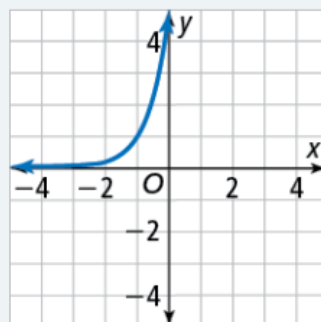
- Find surface wave magnitude of an earthquake with $A = 700$ micrometers, $T = 2$ and $D = 100^\circ$.
- In the formula, $20^\circ < D \leq 160^\circ$. By how much can the size of arc ES affect the surface wave magnitude? Explain.

ASSESSMENT PRACTICE

31. The logarithmic function $g(x) = \ln x$ is transformed to $h(x) = \ln(x + 2) - 1$. Which of the following are true? Select all that apply.

- ☐ $g(x)$ is translated 2 units upward.
- ☐ $g(x)$ is translated 2 units to the right.
- ☐ $g(x)$ is translated 2 units to the left.
- ☐ $g(x)$ is translated 1 unit downward.
- ☐ $g(x)$ is translated 1 unit to the left.
- ☐ The vertical asymptote shifts 2 units to the left.
- ☐ The vertical asymptote shifts 2 units to the right.

32. **SAT/ACT** The graph shows the exponential function $f(x) = 5^{x+1}$. Which of the following functions represents its inverse, $f^{-1}(x)$?



- (A) $f^{-1}(x) = 1 + \log_5 x$
- (B) $f^{-1}(x) = \log_5 x - 1$
- (C) $f^{-1}(x) = \log_5 (x - 1)$
- (D) $f^{-1}(x) = \log_5 (x + 1)$

33. **Performance Task** The logarithmic function $M(d) = 5 \log d + 2$ is used to find the limiting magnitude of a telescope, where d represents the diameter of the lens of the telescope (mm) that is being used for the observation.

Part A Find the limiting magnitude of a telescope having a lens diameter of 40 mm.

Part B Find the equation of the inverse of this function.

Part C Interpret why astronomers may wish to use the inverse of this function. Justify your reasoning.

Part D Using the inverse function, find the diameter of the lens that has a limiting magnitude of 13.5. Check your answer with the table function of your graphing calculator.